

ICE-5121: MACHINE LEARNING



Unsupervised Machine Learning

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- As the name suggests, unsupervised learning is a machine learning technique in which models are not supervised using labeled training dataset. Instead, models itself find the hidden patterns and insights from the given data.
- It can be defined as: *“Unsupervised learning is a type of machine learning in which models are trained using unlabeled dataset and are allowed to act on that data without any supervision.”*
- Unsupervised learning cannot be directly applied to a regression or classification problem because unlike supervised learning. The goal of unsupervised learning is to find the underlying structure of dataset, group that data according to similarities and represent that dataset in a

Why use Unsupervised Learning?

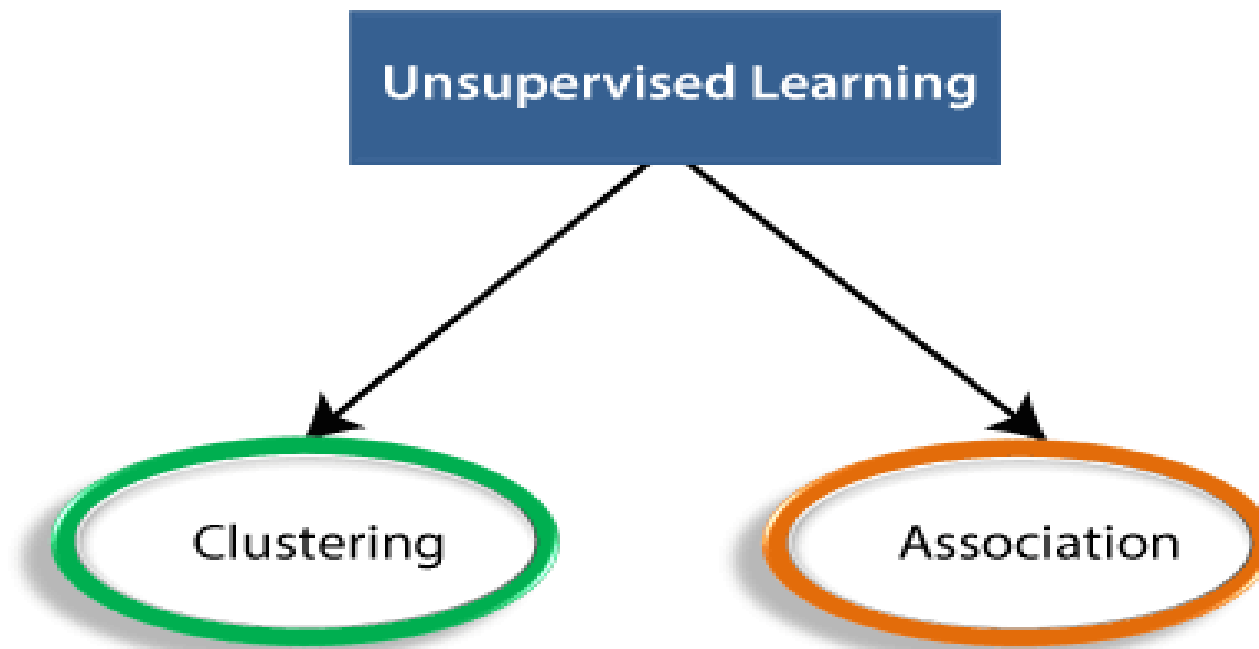
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- Unsupervised learning is helpful for finding useful insights from the data.
- Unsupervised learning is much similar as a human learns to think by their own experiences, which makes it closer to the real AI.
- Unsupervised learning works on unlabeled and uncategorized data which make unsupervised learning more important.
- In real-world, we do not always have input data with the corresponding output so to solve such cases, we need unsupervised learning.

Types of Unsupervised Learning Algorithm

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- The unsupervised learning algorithm can be further categorized into two types of problems:



Types of Unsupervised Learning Algorithm

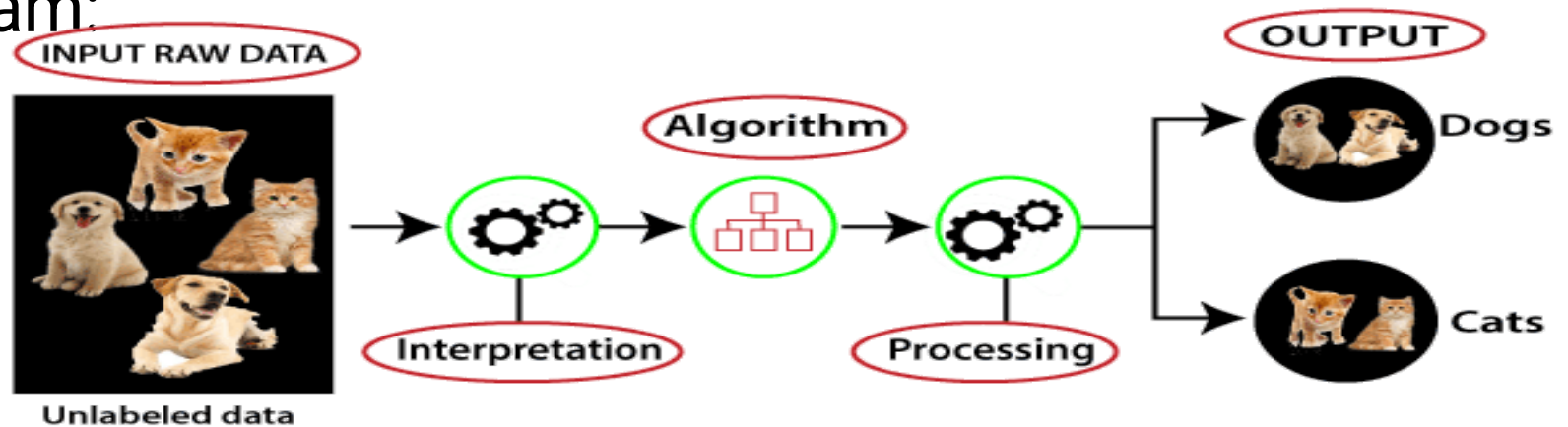
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- **Clustering:** Clustering is a method of grouping the objects into clusters such that objects with most similarities remains into a group and has less or no similarities with the objects of another group.
- **Association:** An association rule is an unsupervised learning method which is used for finding the relationships between variables in the large database. It determines the set of items that occurs together in the dataset. Association rule makes marketing strategy more effective. Such as people who buy X item (suppose a bread) are also tend to purchase Y (Butter/Jam) item. A typical example of Association rule is Market Basket Analysis.

Working of Unsupervised Learning

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Working of unsupervised learning can be understood by the below diagram:



- Firstly, it will interpret the raw data to find the hidden patterns from the data and then will apply suitable algorithms such as k-means clustering, etc.
- Once it applies the suitable algorithm, the algorithm divides the data objects into groups according to the similarities and difference between the objects.

Unsupervised Learning algorithms

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Below is the list of some popular unsupervised learning algorithms:

1. K-means clustering
2. KNN (k-nearest neighbors)
3. Hierarchical clustering
4. Anomaly detection
5. Neural Networks
6. Principle Component Analysis
7. Independent Component Analysis
8. Singular value decomposition

Advantages and Disadvantages of Unsupervised Learning

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Advantages of Unsupervised Learning:

1. Unsupervised learning is used for more complex tasks as compared to supervised learning because, in unsupervised learning, we don't have labeled input data.
2. Unsupervised learning is preferable as it is easy to get unlabeled data in comparison to labeled data.

Disadvantages of Unsupervised Learning:

1. Unsupervised learning is more difficult than supervised learning as it does not have corresponding output.
2. The result of the unsupervised learning algorithm might be less accurate as input data is not labeled, and algorithms do not know the exact output in advance.

Clustering in Machine Learning

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- Clustering or cluster analysis is a machine learning technique, which groups the unlabeled dataset. It can be defined as *"A way of grouping the data points into different clusters, consisting of similar data points. The objects with the possible similarities remain in a group that has less or no similarities with another group."*
- It does it by finding some similar patterns in the unlabeled dataset such as shape, size, color, behavior, etc., and divides them as per the presence and absence of those similar patterns.

Clustering in Machine Learning

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- The clustering technique can be widely used in various tasks. Some most common uses of this technique are:
 - **Market Segmentation**
 - **Statistical data analysis**
 - **Social network analysis**
 - **Image segmentation**
 - **Anomaly detection, etc.**
- Apart from these general usages, it is used by the **Amazon** in its recommendation system to provide the recommendations as per the past search of products. **Netf li** also uses this technique to recommend the movies and web-series to its users as per the watch history.

Clustering in Machine Learning

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- The below diagram explains the working of the clustering algorithm. We can see the different fruits are divided into several groups with similar properties.

